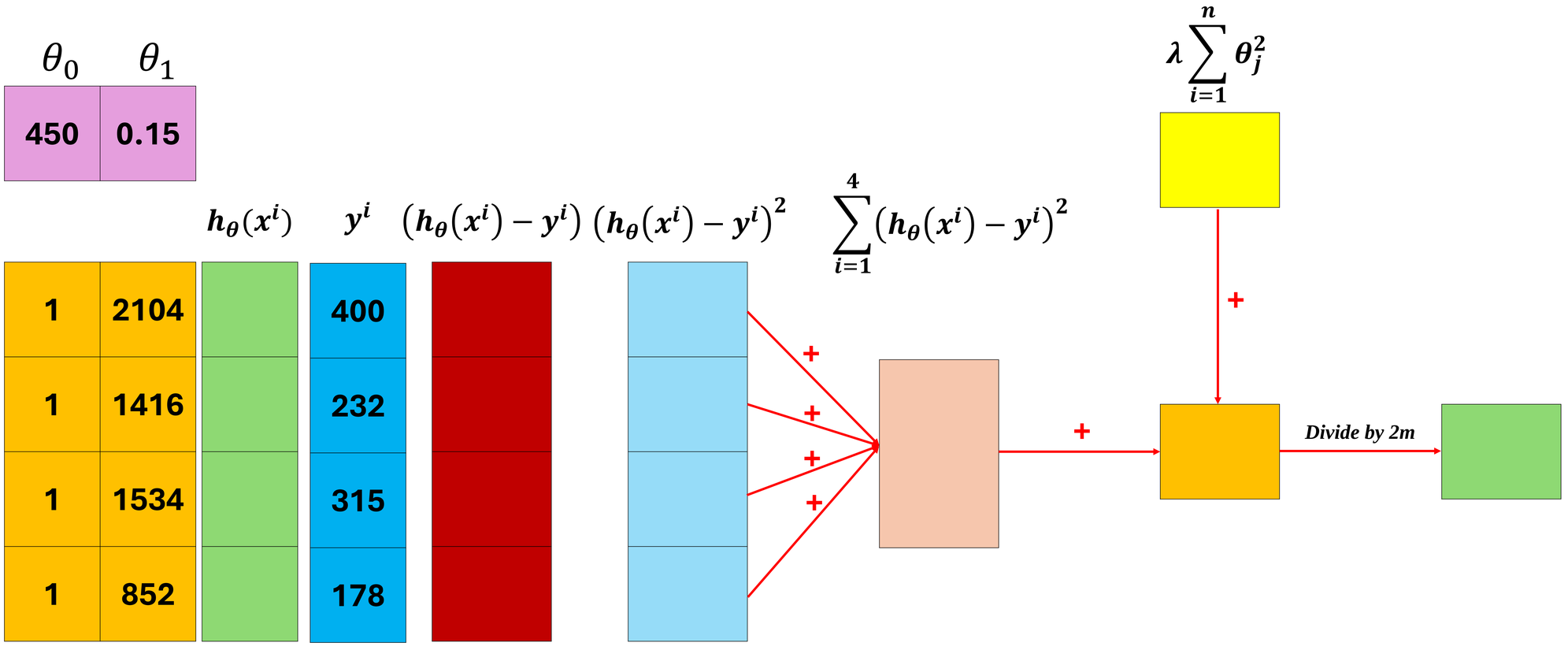
Cost Function of Regularize Linear Regression

$$J(\theta) = \frac{1}{2 \times m}\left[\sum_{i=1}^{m}\left(h_\theta\left(x^i\right) - y^i\right)^2 + \lambda \sum_{j=1}^{n} \theta_j^2\right]$$


θ_0
θ_1
450
0.15
$h_\theta(x^i)$
y^i
$(h_\theta(x^i) - y^i)$
$(h_\theta(x^i) - y^i)^2$
$\sum_{i=1}^{4}(h_\theta(x^i) - y^i)^2$
$\lambda \sum_{i=1}^{n} \theta_j^2$
1
2104
400
1
1416
232
1
1534
315
1
852
178
+
+
+
+
+
+
+
Divide by 2m

Linear Regression Regularized Gradient Descent workbook

